

COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Midsemester Exam #1

Only one sheet of paper 5 x 7 is allowed

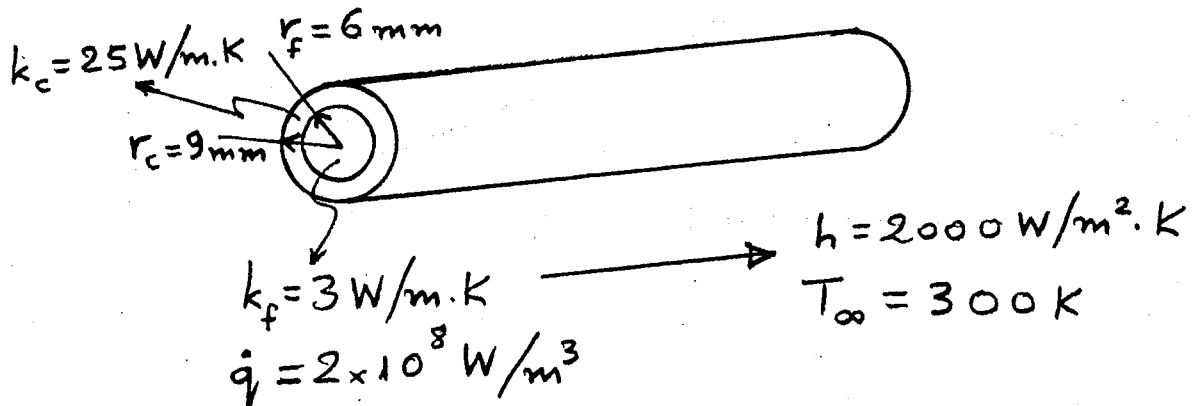
Duration: 50 minutes --- February 10, 2020

Both Problems must be done

Problem 1: (50 points)

A nuclear reactor fuel element consists of a solid cylindrical pin of radius r_f and thermal conductivity k_f . The fuel pin is in good contact with a cladding material of outer radius r_c and thermal conductivity k_c . Consider steady-state conditions for which uniform heat generation occurs within the fuel at a volumetric rate $\dot{q}$ and the outer surface of the cladding is exposed to a coolant that is characterized by a temperature T_∞ and a convection coefficient h .

- Obtain equations for the temperature distributions $T_f(r)$ and $T_c(r)$ in the fuel and cladding, respectively. Express your results exclusively in terms of the foregoing variables.
- Consider a uranium oxide fuel pin for which $k_f = 3 \text{ W/m}\cdot\text{K}$ and $r_f = 6 \text{ mm}$ and cladding for which $k_c = 25 \text{ W/m}\cdot\text{K}$ and $r_c = 9 \text{ mm}$. If $\dot{q} = 2 \times 10^8 \text{ W/m}^3$, $h = 2000 \text{ W/m}^2\cdot\text{K}$, and $T_\infty = 300 \text{ K}$, what is the maximum temperature in the fuel element?



Problem 2: (50 points)

A solid cone as shown is of circular cross section and the radius of its base is 0.5 m. The cone of length 1 m and thermal conductivity $20 \text{ W/m}\cdot\text{K}$ has uniform volumetric heat generation of 100 W/m^3 . A constant heat flux of 0.1 W/m^2 is leaving the cone through the lateral surface of the cone, while the end surface at $x = 0$ is maintained at $T_0 = 30^\circ\text{C}$. Determine the temperature profile $T(x)$ and the temperature $T(L)$.

